Component Kit

EE5203 Microprocessors — what to buy for the semester

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Buy this once, in the first week. You need it for homework and for practising before each laboratory session.

Item	Quantity	Notes
ST Nucleo-F401RE board	1	Nucleo-64, STM32F401RE. Not the F411, not a "Blue Pill".
USB Mini-B data cable	1	Mini-B, not Micro-B and not USB-C. Must carry data, not only power.
Solderless breadboard	1	Half-size, about 400 tie points.
Jumper wires, 40-way	1 set of each	Male/male, male/female, female/female. Swap a suspect wire before you debug the code.
LEDs, any colour	10	5 mm.
Resistors, 330 Ω	10	LED current limiting.
Resistors, 10 kΩ	2	Pull-ups and dividers.
Push buttons, 2-pin	5	Two-pin parts sit on a breadboard more reliably than four-pin ones.
Potentiometer, 10 kΩ	2	Linear taper. The analogue knob from Lab 9.
Digital multimeter	1	Nothing fancy. The cheapest one that reads DC volts, continuity and resistance is fine.

Order the **board and cable first**. They are the only items you cannot substitute, and the most likely to arrive late.

Labs 1 to 7 need the board alone; LD2 and B1 are already on it. Breadboard wiring starts at Lab 8.

Two mistakes that cost a week

- **A charge-only USB cable.** It powers the board but carries no data, so the programmer never sees it. The most common cause of "my board does not work".
- **The wrong plug.** Micro-B and USB-C will not fit the Nucleo. Check the photograph of the connector before ordering.

An assorted resistor pack covers both resistor values for about the same price. Come and talk to me if cost is the obstacle, rather than arriving without the parts.